SymK Reader Orientation and Shared Glossary

A plain-language map of the Foundation Paper set

SymK is a general framework and engineering discipline for responsible Knowledge Engineering through governed cooperation among different forms and Bearers of Intelligence. It calls us to engineer the conditions in which difficult questions can produce scoped, evidence-grounded, challengeable and revisable knowledge work - while never confusing information with Knowledge, performance with Intelligence, capability with Authority, cooperation with legitimacy, or a successful outcome with proof.

The principal practical result SymK is intended to enable is a class of governed, domain-centred, knowledge-intensive systems - principally Domain Intelligence Amplifiers - in which qualified people, AI and other participants cooperate through governed domain Knowledge to answer complex questions. LexBrain is one concrete derived-project example: a legal-domain knowledge system shaped by SymK, not SymK itself.

SymK

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> **Review status:** Governed v0.3 companion review draft. It is coordinated with the four v0.3 Foundation Papers but does not replace them or complete Human review.

Welcome

SymK asks a demanding but practical question: how can people and other capable systems work together on knowledge-intensive problems without confusing fluent output with truth, capability with authority, participation with responsibility, or change with improvement?

The project began around symbiotic cooperation between Human and Artificial Intelligence. It developed Domain Intelligence Amplifiers - often shortened to DIA - as a principal way of putting that cooperation to work in a bounded domain. SymK later broadened into a general framework and engineering discipline for responsible Knowledge Engineering through governed cooperation among different forms and Bearers of Intelligence.

This companion gives you the small amount of shared orientation needed to move comfortably among four Foundation Papers:

1. *Knowledge as the Shared Epistemic Medium*;
2. *Intelligence Beyond the Human*;
3. *Formation, Education, and Training in SymK*; and
4. *Engineering Epistemic Conditions*.

You do not need to read this companion before those papers. Each paper is designed to stand alone. Use this document when you want a common map, a recurring example or a quick explanation of a term.

Knowledge Engineering, before SymK

In ordinary language, Knowledge Engineering is the disciplined work of determining how knowledge in a particular Domain is gathered, expressed, organized, connected to evidence, reviewed, maintained, challenged and made usable by people and computational systems. Its artifacts may include conceptual models, controlled vocabularies, ontologies, knowledge bases, reasoning structures, provenance records, validation rules and the processes that keep them useful. It may draw on knowledge management, data engineering, software engineering and AI engineering, but it is not identical to any one of them.

SymK retains that practical concern but extends the engineering object beyond representations and technical artifacts. It treats the full epistemic arrangement as designable and governable: participants, Questions, claims, Grounds, Evidence, roles, Responsibilities, Authorities, Representations, decisions, consequences, feedback and Correction. SymK engineers conditions for responsible knowledge work; it does not manufacture truth or Knowledge.

This gives the papers different but connected roles:

- **Knowledge Engineering is the integrative center:** it brings the other concepts into the design, construction, governance, evaluation and evolution of epistemic conditions.
- **Knowledge is constitutive:** it names the scoped epistemic achievement and shared medium that the engineering must not reduce to information, storage, consensus or approval.
- **Intelligence is constitutive:** it names the capacities and qualified Bearers that participate in the work without acquiring Authority merely from capability.
- **Formation is constitutive:** it makes comparatively durable change in people, systems and organizations visible across time.
- **Education and Training are conditional formative practices:** they matter when deliberately shaping participants or capabilities, but they are not universal requirements of every SymK undertaking.

This is an architecture of purpose, not a hierarchy of constitutional rank. Each accepted concept retains its own meaning and no paper may silently redefine another. The architecture also remains open: future supporting concepts may be admitted through evidence-controlled governance when they protect a necessary distinction or relationship. Usefulness or implementation convenience alone does not make a new concept Foundational.

What SymK is - and is not

In plain language, SymK is a way to design and govern cooperative knowledge work. It helps participants make visible:

- what question they are trying to answer;
- which people or systems contribute and in what roles;
- what is claimed and what supports or challenges the claim;
- where the claim applies and what uncertainty remains;
- who is responsible for which part of the work;
- who is competent to authorize a decision or use;
- what consequences follow; and
- how correction and learning return to earlier steps.

SymK is not a promise that cooperation is always good. It is not a claim that artificial systems are equivalent to Humans. It is not one universal ontology, workflow, database or software platform. It does not turn Evidence into authority, capability into permission, consensus into truth, or a successful outcome into proof that every earlier step was correct.

The name “SymK” remains the controlled historical proper name through phase 3.0. The project has deliberately postponed the question of whether that name still fits until after accepted 3.0E. That future challenge does not alter the meaning or status of the current papers.

The four papers as one conversation

The papers are easier to understand when treated as four answers to connected questions.

Knowledge: What may responsibly count as Knowledge, and how is it kept distinct from a claim, record, representation, approval or authorized use?

Intelligence: What kind of capacity allows an organized system to interpret or respond across varying situations, and what evidence supports an attribution of that capacity?

Formation, Education and Training: How do people, systems and organizations change over time, and how do we avoid calling every change Learning, successful Training, Education or improvement?

Knowledge Engineering: What can be designed when truth and Knowledge cannot simply be manufactured? SymK's answer is: design and govern the epistemic conditions around inquiry, evidence, representation, review, reliance, consequence and correction.

The papers are mutually necessary but architecturally asymmetric. Knowledge Engineering is the integrative center of the set: it brings the other concepts into the design and governance of epistemic conditions. The Knowledge and Intelligence papers define achievements and capacities that this engineering must not distort. The Formation paper explains change over time; Education and Training are conditional formative practices rather than universal requirements. This centrality concerns SymK's purpose, not constitutional rank, and no paper may silently redefine concepts governed by another.

A running example

Imagine that a bridge-inspection team receives sensor readings, photographs, an engineer's assessment and an AI-generated warning about a crack.

The **Knowledge** question is not whether the warning has been stored. It is what is actually known, by which Bearer, about what, within which Scope, on what Grounds and with what possible defeaters.

The **Intelligence** question is not whether the AI gave one impressive answer. It is what capacity the Human, artificial, collective or hybrid Bearer has shown across materially varying conditions. A performance is evidence about capacity, not the entire capacity.

The **Formation** question appears when the team changes its routines, practises a new inspection method or updates the model. Those activities create formative conditions. Evidence is still needed to establish what changed, in whom or what, whether it endured, and whether it was beneficial.

The **Knowledge Engineering** question concerns the whole arrangement: instruments, calibration, participant roles, claim and evidence records, representations, model provenance, review, challenge, competent operational authority, reliance decisions, consequence monitoring and correction.

Suppose the evidence suggests an urgent risk. SymK does not invent an emergency containment duty or authority. The evidence triggers urgent consideration by the competent operational or external authority. Evidence may challenge the premises, applicability, consequences or decisions made under authority; change to an authority or governed artifact follows the competent procedure.

The shared lifecycle

The Foundation Papers often refer to a branching, recursive lifecycle. It can be read in ordinary language:

1. a situation is framed as one or more Questions;
2. participants gather Evidence and contribute Reasoning;
3. one or more Answers or candidate claims are produced;
4. their quality, cost, timeliness, uncertainty and Value are assessed;
5. competent authority may authorize a decision or reliance;

6. action or non-action has consequences;
7. feedback, challenge, response and Correction may return to any earlier point; and
8. comparatively durable change may become Learning or another form of Formation.

The sequence is not an automatic pipeline. A question can branch. Evidence may change the framing. A decision may reject an answer. Consequences may reveal that a claim, method, authorization or policy must be reconsidered. A favorable outcome does not prove that the process was epistemically sound, and an unfavorable outcome does not by itself prove the opposite.

Three Views

SymK uses three coordinated Views because no single description is sufficient.

The Human View asks what people experience, understand, can challenge and may be affected by. It protects Human legibility without treating Humans as the definition of every Intelligence.

The Scientific View asks what evidence, method, uncertainty, alternative explanations and limits support a claim. It does not become the sole source of legal, ethical or organizational authority.

The Engineering View asks what must be represented, built, tested, operated, monitored and corrected. An implementation does not silently redefine the concepts it implements.

The Views should inform one another. None may quietly replace the others.

Shared glossary

These explanations describe how the current Foundation Papers use recurring terms. They are a reading aid, not the future mature vocabulary, contract package or a constitutional amendment.

Affected-subject standing/challenge locus - the place in the analysis where the experience, claims and ability to challenge of materially affected subjects must be represented and routed. Its precise classification remains unresolved; it is not a settled twelfth Authority kind and does not automatically create veto, legal standing, participation or remedy.

Amplification - a scoped comparison in which a capability or result increases relative to a declared baseline. More output is not automatically more Knowledge, Value or responsible performance.

Answer - a response related to a Question. It may be partial, conditional, contested or wrong. Being an Answer does not establish quality or truth.

Authority - competence to make a particular kind of decision within a jurisdiction. SymK distinguishes eleven typed Authority kinds in the current analytical baseline; evidence, Intelligence, Knowledge or affectedness alone does not manufacture Authority.

Bearer - the identified subject or organized system to which a property such as Knowledge, Intelligence or Learning is attributed. The Bearer may be Human, artificial, collective, institutional or hybrid, but the boundary must be justified.

Context - circumstances relevant to interpreting a claim, attribution or decision. Context does not excuse leaving the applicable boundary unstated.

Cooperation - a thin relational process in which contributions become mutually relevant under some coordination. Cooperation is not automatically beneficial, voluntary, legitimate, successful or sufficient.

Domain - a bounded field of concern, practice or inquiry in which terms, evidence, competence and authority may have specific meanings.

Domain Intelligence Amplifier (DIA) - a principal SymK realization orientation for amplifying responsible intelligence in a declared Domain. It is important but does not exhaust SymK.

Education - broad, developmental, integrated, participatory and challengeable Formation that connects Knowledge, judgment, conduct, relationship and the capacity to examine and revise the frame.

Epistemic conditions - the participants, processes, roles, resources, Representations, controls, environments and feedback that enable, constrain or impair responsible Knowledge work.

Evidence - material that bears on a claim, premise, attribution, applicability, consequence or decision. Evidence may challenge what is done under authority; competent procedure governs changes to the authority or governed artifact itself.

Formation - temporally extended shaping or reorganization of an identified subject's comparatively durable capacities, dispositions, practices, judgments, relationships or organized patterns.

Grounds - the reasons, basis or support offered for a claim or attribution. Grounds may be inadequate, defeated or outside Scope.

Intelligence - a substrate-realized, multidimensional capacity of an organized system to integrate relevant resources in producing context-sensitive interpretations or effective responses across materially varying situations.

Knowledge - a scoped epistemic achievement attributable to a Bearer, appropriately connected to truth, competence or encounter rather than to mere luck, assertion, acceptance or Representation.

Knowledge Engineering - the disciplined work of making Domain knowledge expressible, organized, evidence-connected, reviewable, maintainable, challengeable and usable. In SymK, its direct object extends to the complete epistemic conditions around that work. It does not manufacture truth or Knowledge directly.

Learning - comparatively durable, experience-linked change at an identified Learner level. Component change does not automatically establish system Learning.

Projection - a selective transformation or view, such as a score, class, chart or summary. Its purpose, source and material loss should remain visible.

Question - a governed inquiry object that frames what is being asked without forcing a single answer or path.

Reasoning - a contribution or process connecting questions, premises, evidence and candidate answers. A recorded chain can still be unsound or incomplete.

Representation - an expression of something for a purpose. A document, graph, model, statement or record is not identical to reality, Knowledge or the represented subject.

Responsibility - a sourced, scoped, phased and challengeable relationship-family concerning what a subject must do, answer for or respond to. It is not manufactured by causal contribution or capability alone.

Scope - the declared boundary within which a claim, attribution, method or authorization is intended to apply.

Supporting concept - a governed distinction needed to preserve a material relation, boundary or failure mode without independently organizing SymK's identity. New supporting concepts may be admitted through evidence-controlled governance; they do not become Foundational by convenience or repeated implementation use.

Training - directed and structured formative activity toward declared performance or disposition conditions. Completion does not prove success or Education.

Value - a plural and scoped assessment of what matters for a purpose or affected subject. Value is not one universal score and does not erase constraints, harms or distribution.

Authority, evidence and urgency

Three safeguards recur because their collapse can cause serious misunderstanding.

First, affected-subject standing and challenge must be visible, but their exact authority classification remains unresolved. The current papers therefore treat this as a standing/challenge locus distinct from the typed Authority kinds.

Second, Evidence does not “challenge authority” as though information automatically changed a mandate, office, rule or artifact. It challenges claims, premises, applicability, consequences or decisions made under authority. Any formal change then follows the competent procedure.

Third, urgent evidence does not create a universal emergency duty or containment power. It triggers prompt consideration by the competent operational or external authority. Domain law, policy, role and circumstance determine what that authority may or must do.

How to read deeply without getting lost

When a paragraph becomes technical, ask five questions:

1. What is the paper distinguishing from what?
2. What proposed Bearer, Scope and Context are being discussed?
3. Is the text describing reality, a claim, Evidence, a representation, an assessment, an authorization or a consequence?
4. Which View is speaking, and has another View been silently substituted?
5. What does the passage explicitly refuse to imply?

The detailed papers use precision because seemingly small shortcuts can create authority, responsibility, truth or value claims that the evidence does not support. You do not need to memorize the vocabulary. Follow the distinctions and return to this glossary when necessary.

Status and limits

Version: 0.3 coordinated Human-review draft
Date: 24 August 2026
Baseline: `2.1-rc.1`; Stage-Accepted `SYMK-2X-DR-009` through `SYMK-2X-DR-018`
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Review lineage: Stream H Human finding corrections `SYMK-2X-EV-173` through `SYMK-2X-EV-176`; coordinated v0.3 designation `SYMK-2X-EV-177`
Authority: explanatory companion only; not a constitutional norm, accepted definition source, final vocabulary, contract, ontology, workflow or implementation specification
Pending: complete Human reread, genuine-newcomer testing, any assistive-technology evaluation actually used, and publication acceptance

If this companion and an exact Stage-Accepted decision differ, the decision controls. If understanding a Foundation Paper’s main argument requires this companion, that paper still needs revision.